Firm Size and the Great Recession

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Abstract

The US firm-size distribution has changed substantially over time, with increasingly more large firms and fewer small firms. How does this change affect the economy's response to a financial crisis? I take a dynamic stochastic general equilibrium model of heterogeneous firms that choose their capital and debt subject to a borrowing constraint, modified to let firms choose what they use for collateral, and examine the recovery from a financial crisis when firms have a choice of whether or not to borrow against capital or earnings. I compare the economy's response for the firm-size distribution that prevailed at the start of the 2007 Financial Crisis to the one that prevailed in 1980. I find that the trough of the recession is shallower under the 1980 firm-size distribution due to the higher proportion of smaller firms compared to 2007. These small firms increase their capital and output to ameliorate the effect of the tighter collateral constraints. I also find that the majority of the recession is driven by the tightening of the capital collateral constraint, indicating that a capital constraint alone might be sufficient to model the Great Recession.

Key words: Financial crisis, firm-size distribution, collateral constraints, financial frictions

JEL Codes: E32, E20, E44

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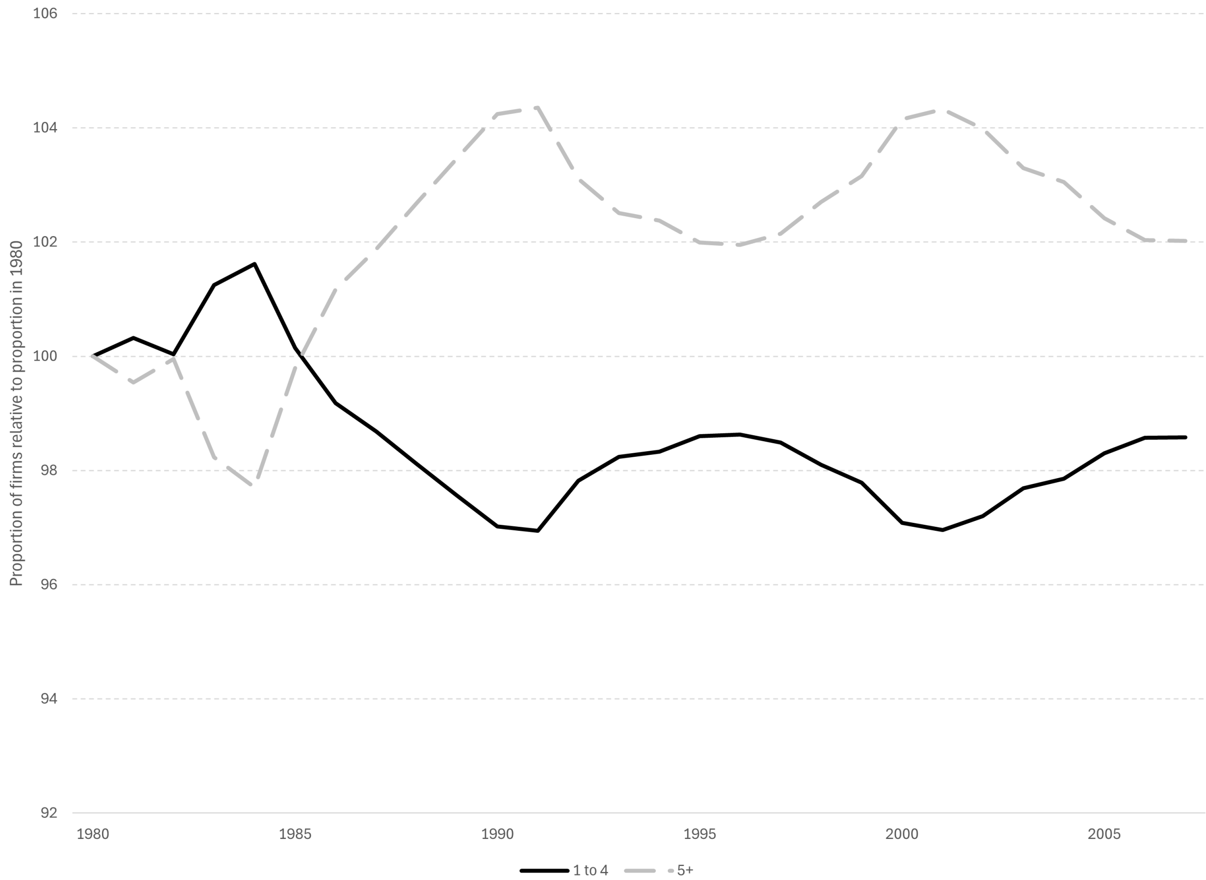
1. Introduction

The distribution of firms over size has changed substantially over time in the USA. In particular, the proportion of large firms has increased while the proportion of firms that are small has decreased. For example, Figure 1 shows that, relative to 1980, the proportion of firms with between one and four employees (the solid black line) decreased by roughly two percentage points between 1980 and 2007, while the proportion of firms with five or more employees increased by roughly the same amount.[1]

Figure 1: Proportion of firms by number of employees relative to 1980



Note: Using BEA firm-size distribution data for each year between 1980 and 2007. The solid black line shows the proportion of firms with between one to four employees relative to the proportion of such firms in 1980, whereas the dashed grey line shows the proportion of firms with five or more employees (also relative to the proportion of such firms in 1980).

As smaller firms are more likely to be financially constrained (see, for example, Beck, Demirguc-Kunt and Maksimovic (2005)), this means that the change in the firm-size distribution could have affected the economy's response to the 2007 / 2008 financial crisis and recovery from it. In particular, the fact that firms were less likely to be small in 2007

[1] The picture is very similar using different cut-offs for the number of employees. For example, using the same grouping of small firms having 1-19 employees and large firms having 20+ employees as used by Fort et al. (2013), the employment by small firms decreases by roughly one percentage point over the same period whereas the employment by large firms increased by about eight percentage points.

relative to 1980 could imply that the Great Recession was not as severe as it would have been had the 1980 firm-size distribution been in place when the financial crisis occurred. Conversely, Luttmer (2011)'s finding that smaller firms tend to grow more quickly than larger firms combined with the fact that firms tended to be larger in 2007 could imply that the recovery from the Great Recession was slower than it would have been had the firm-size distribution been the one that prevailed in 1980.

To investigate which of these potential factors affecting the economy's response to the Great Recession, I use a standard dynamic stochastic general equilibrium model of heterogeneous firms (modified to let firms choose the type of collateral against which they borrow) to examine the effect of changes in the firm-size distribution on the recovery from a financial crisis. Specifically, I examine the impact of an economy's recovery when the firm-size distribution is as it was at the start of the Great Recession (i.e. in 2007) and compare it against a recovery from the same shocks if the firm-size distribution at the start of the Great Recession had reflected that which prevailed in 1980 (i.e. with relatively more small firms and relatively fewer large firms).

Firms invest in capital and take on debt subject to a borrowing constraint, where firms can choose whether that borrowing constraint depends on their (future) stock of capital or their expected future earnings. Letting firms choose their type of collateral reflects recent findings by Lian and Ma (2021) that more firms borrow against earnings than against assets despite firms having a free choice as to what they use for collateral, and Shah (2024)'s result that the type of collateral against which firms borrow affects the economy's recovery from a financial shock.

The recession stemming from a financial crisis is smaller under the 1980 firm-size distribution compared to the 2007 one. Specifically, the trough in each of output, consumption, and the capital stock is roughly 0.5% points closer to the steady-state level (i.e. the trough is not as deep) when the firm-size distribution matches that which prevailed in 1980. This is due to the larger proportion of smaller firms: when hit with a financial crisis, small (and medium-sized) firms actually increase their output and capital stock relative to the steady-state to try to ameliorate the tightening of the collateral constraint, as found by Shah (2024). Hence, the 1980 economy with its higher proportion of smaller firms benefits from this effect more than does the 2007 economy, such that the recession is less severe in the former firm-size distribution.

Moreover, I find that the majority of the effect of the recession stems from the tightening of the capital collateral constraint rather than the tightening of the earnings collateral constraint. Simulating a financial crisis in which one of the constraints is tightened without the other being tightened and comparing that to the full financial crisis in which each constraint is tightened consecutively shows that tightening the capital collateral constraint accounts for more than 80% of the ultimate trough of output, consumption, and the capital stock. As such, modelling a financial crisis using solely a capital collateral constraint would capture the vast majority of the economy's path during and after a financial crisis.

I contribute to two main strands of literature. First, by examining the effect of the change in the firm-size distribution on the response to financial shocks, I add to the literature showing the importance of taking the firm-size distribution into account. In particular, Jo (2021) shows that accurately capturing the Pareto nature of the firm-size distribution makes a substantial difference in a model economy's recovery from a recession caused by a financial shock, and my results reinforce these by showing that even small changes in the firm-size distribution can have sizeable effects on the recovery from a financial crisis. My results also provide an alternative perspective to Cabral and Mata (2003)'s findings that smaller firms tend to be more financially constrained and that, therefore, the firm-size distribution shifting towards there being more smaller firms can lead to a financial crisis having a bigger effect: specifically, I find that the presence of more small firms can help to reduce the size of a recession caused by a financial crisis as they grow more quickly (which is also consistent with Fort et al. (2013)'s finding that smaller firms have larger magnitude responses to shocks than do larger firms). In addition, my finding that smaller firms are affected more during a financial shock is consistent with Duygan-Bump, Levkov and Montoriol-Garriga (2015)'s results that workers at smaller firms were more likely to be laid off than workers at larger firms.

This also matches the aforementioned Beck, Demirguc-Kunt and Maksimovic (2005)'s results that smaller firms are the most constrained, such that any tightening of the financial constraints they face hinders their growth such that small firms will react so as to reduce the impact of the tighter financial constraints. Finally, my results lend evidence for Luttmer (2011)'s finding that factors that hamper the growth of smaller (and younger) firms have sizeable effects on the aggregate economy because those smaller (younger) firms are the ones that grow the quickest.

The second strand of literature to which I contribute is that regarding the importance of the choice of collateral against which firms borrow. By allowing firms to choose whether they borrow against earnings or against assets (their capital stock) Lian and Ma (2021)'s finding that roughly 80% of firms borrow against their earnings rather than against any assets. Each of Shah (2024), Drechsel (2023), and Choi (2022) previously examined the effect of firms borrowing against different types of collateral by exogenously imposing that firms were only borrowing against one type of collateral. In contrast to those papers, I allow firms to choose whether they borrow against their earnings or their assets, which allows me to observe an endogenous response of firms' choice of collateral to changing financial conditions when the constraints for different types of collateral tighten independently.

Moreover, letting firms choose their collateral type means I can also incorporate Ingholt

(2022) and Kariya (2022)'s results that the constraints faced by each type of collateral borrower tightened differently at different times during the Great Recession and examine the extent to which this additional element gets closer to matching the path of the slow recovery from the Great Recession. As such, I also add to the wealth of research, such as Khan and Thomas (2013), Jermann and Quadrini (2012), and Kiyotaki and Moore (1997) examining the recovery from a financial crisis

The remainder of this paper proceeds as follows: Section 2 sets out the model I use to investigate the effect of different types of collateral, while Section 3 provides an analysis of firms' optimum choices. Section 4 contains a description of the calibration of my model, and Section 5 presents the responses of the model economy to a financial crisis in the form of tightening each collateral constraint. Section 6 examines the relative contribution of tightening each collateral constraint and Section 7 concludes.

2. Model

In this Section I describe the features of the discrete-time stochastic general equilibrium model. There are two types of agents: a mass of heterogeneous firms that produce the single consumption good; and an infinitely-lived representative household that owns the firms (receiving any profits from them) and obtains utility from consumption C and leisure $1 - N$.[2]

A firm is characterised by the combination of their exogenous idiosyncratic output productivity ε, their stock of physical capital k (which depreciates at rate δ), and their stock of debt b. The distribution of firms over these three variables represented by μ. I combine a firm's physical capital and debt into a firm's "cash-on-hand" at the start of a period according to Jo (2021). In addition, firms face an exogenous probability π_d of exiting each period.

All firms produce output y using their physical capital stock and labour ℓ. If a firm survives (i.e. is not hit with the exogenous death shock) then it gets to choose its physical capital and debt for the next period. When choosing its debt for the next period, a firm also decides whether it uses its capital or its earnings as collateral: this choice will affect the amount that a firm can borrow. Specifically, firms face a collateral constraint where they can borrow up to θ_k against their capital stock or up to θ_e against their earnings, and can choose which of those constraints it faces.

Finally, firms are subject to aggregate uncertainty in the form of shocks to aggregate total factor productivity z as well as to each collateral constraint (i.e. to θ_k and to θ_e). For

[2] Households also have access to a full set of state-contingent claims, but as the household is representative in my model, these are in zero net supply in equilibrium. As such, I do not explicitly model these claims here for convenience.

ease of notation, these exogenous state variables are represented by $S = (z, \theta_k, \theta_e)$. There is also a mass of entrants that enter each period with zero debt and physical capital k^N (which is some fraction χ of the aggregate capital stock).[3]

I present the optimisation problems of each type of agent in turn.

2.1. Households

The representative household obtains utility from consumption C and leisure $1 - N$ (with N representing the household's labour supply). The household discounts the future at its rate of time preference β and earns wages $w(S, \mu)$ by supplying labour to firms. The household owns shares λ in the firms (where the ex-dividend price of shares in a firm with capital stock k, debt b, and exogenous output productivity ε next period is denoted by $\rho_1(k', b', \varepsilon'; S, \mu)$ and the price inclusive of the dividend being $\rho_0(k, b, \varepsilon; S, \mu)$) and can borrow or save in one-period discount bonds ϕ with price $q(S, \mu)$.

As such, households solve the following problem:

$$V^h(\lambda, \phi; S, \mu) = \max_{C,N,\phi',\lambda'} U(C, 1 - N) + \beta E_{\pi_S} V^h(\lambda', \phi'; S', \mu') \tag{1}$$

$$\begin{aligned} C + q(S,\mu)\phi' + \int \rho_1(k', b', \varepsilon'; S, \mu)\lambda'(d[k' \times b' \times \varepsilon')] \\ = w(S,\mu)N + \phi + \int \rho_0(k, b, \varepsilon; S, \mu)\lambda(d[k \times b \times \varepsilon) \\ \mu' = \Gamma(S, \mu) \end{aligned} \tag{2}$$

where Equation (2) is the household budget constraint. The distribution of firms moves according to the law of motion Γ.

The optimum conditions that result from this problem determine the wage rate and the bond price (loan discount factor). Specifically, the wage $w(S, \mu) = \frac{D_2 U(C,1-N)}{D_1 U(C,1-N)}$ is given by the marginal rates of substitution between consumption and leisure for the household, while the bond price $q(S, \mu) = \beta \frac{E_{\pi_{S'}} D_1 U(C'_{S'}, 1-N'_{S'})}{D_1 U(C,1-N)}$ is given by the inverse of the (expected) gross real interest rate. Furthermore, as per Khan and Thomas (2013), this problem determines the firm's stochastic discount factor (as the household earns all firms). As such, the firm problem incorporates this by scaling any current period profit by $p(S, \mu) = D_1 U(C, 1 - N)$.

The specific form of the household utility function takes Rogerson-Hansen preferences such that $U(C, 1 - N) = \log C + \varphi(1 - N)$, with φ representing the household's disutility of working.

[3] As the aggregate capital stock changes over time, this means that the capital stock with which entrants begin the period also changes over time.

2.2. Firms

At the start of each period, firms learn their idiosyncratic output productivity (which is persistent via a one-period Markov Chain) and whether or not they are subject to the exogenous death shock.[4] After observing these shocks, firms produce output y using their stock of physical capital and hiring labour in a Cobb-Douglas production function weights α on capital and γ on labour. The idiosyncratic productivity allows me to capture the firm-size distribution, thereby enabling my examination of the effect of this distribution on the recovery from a financial crisis.

Complete reversibility of capital in my model means that, as per Jo (2021), a firm can be characterised according to its cash-on-hand at the start of the period: specifically, a firm's cash-on-hand is given by $m = \max_{\ell} z\varepsilon k^{\alpha}\ell^{\gamma} - w(S,\mu)\ell + (1-\delta)k - b$.

After producing, firms that were not subject to the exogenous death shock decide their capital stock k' and debt b' subject to a collateral constraint and the firm's profits being non-negative. Firms also choose what they use as their collateral, which affects how much they can borrow: they can use their future capital stock as collateral in which case they can borrow up to a multiple θ_k of that capital stock; or they can borrow against their expected future earnings, in which case they can borrow up to a multiple θ_e of those expected future earnings.[5] [6] I denote this choice of collateral as o. Any firm that received the exogenous death shock, instead, pays off their current debts, sells their undepreciated capital, and pays out any remaining amounts as dividends to households.

Let V represent the value of an incumbent firm at the start of the period and V^1 be the value of a firm that does not receive the exogenous death shock. The incumbent firm's optimisation problem can be defined recursively as

$$\begin{aligned} V(m,\varepsilon;S,\mu) = \pi_d p(S,\mu)m \\ + (1-\pi_d)V^1(m,\varepsilon;S,\mu) \end{aligned} \tag{3}$$

$$V^1(m,\varepsilon;S,\mu) = \max_{o}(V^k, V^e) \tag{4}$$

where V^k is the value obtained by a firm that chooses to use its future capital stock as collateral and is given by

[4] This exogenous probability of death each period is necessary in to ensure that the model economy is not entirely populated by large firms. Without the exogenous death shock, all firms would survive until they reached their optimum size, such that the steady-state would consist only of large firms.

[5] I use expected future, rather than current, earnings to ensure that both specifications of the constraint reflect a lender's desire to be able to recoup any debt in the next period, and (expected) future earnings better reflects the ability of lenders to recoup those debts than do current earnings.

[6] Throughout this model, I keep the price of capital (relative to the price of the consumption good) constant and equal to one. As such, I abstract from the potential impact of fluctuations in the relative price of capital when comparing the effect of different forms of collateral constraints.

$$V^k(m,\varepsilon;S,\mu) = \max_{k',b',\ell} p(S,\mu)D + \beta E_{\pi_S} E_{\pi_\varepsilon} V(m',\varepsilon';S',\mu') \tag{5}$$

subject to

$$\begin{aligned}
0 \leq D &= m - k' + q(S,\mu)b' \\
b' &\leq \theta_k k' \\
\mu' &= \Gamma(\mu) \\
m' &= m(k',B',\varepsilon';S',\mu') \\
&= \max_{\ell'} z\varepsilon k'^{\alpha}\ell'^{\gamma} - w(S',\mu')\ell' + (1-\delta)k' - b'
\end{aligned} \tag{6}$$

while V^e is the value obtained by a firm that uses its expected future earnings as collateral and is given by

$$V^e(m,\varepsilon;S,\mu) = \max_{k',b',\ell} p(S,\mu)D + \beta E_{\pi_S} E_{\pi_\varepsilon} V(m',\varepsilon';S',\mu') \tag{7}$$

subject to

$$\begin{aligned}
0 \leq D &= m - k' + q(S,\mu)b' \\
b' &\leq \theta_e E_{\pi_S} E_{\pi_\varepsilon} (\max_{\ell'} z'\varepsilon' k'^{\alpha}\ell'^{\gamma} - w(S',\mu')\ell') \\
\mu' &= \Gamma(\mu) \\
m' &= m(k',B',\varepsilon';S',\mu') \\
&= \max_{\ell'} z'\varepsilon' k'^{\alpha}\ell'^{\gamma} - w(S',\mu')\ell' + (1-\delta)k' - b'
\end{aligned} \tag{8}$$

The only difference between Equations (5) - (6) and Equations (7) - (8) is the form of the collateral constraint: in the former, the collateral constraint has firms using their future capital stock as collateral, while in the latter, firms use their expected future earnings as collateral. Finally, there is also a mass of entrants that matches the number of exiting firms each period; these entrants come in with zero debt and a capital stock that is a fraction χ of the aggregate stock of capital.[7]

2.3. Equilibrium

Let $\mathbb{I}^O$ represent those firms in the distribution that are continuing incumbent firms and $\mathbb{I}^{\mathbb{N}}$ denote entrants, a Recursive Competitive Equilibrium is a set of prices p, w, q, ρ_0, ρ_1,

[7] For simplicity half of entrants are allocated to having borrowed against capital and half are allocated to having borrowed against earnings when they enter. The fact that their debt is zero upon entry means that this does not have a substantial effect on the model's results.

and functions for quantities k', l, b', o, C, N,ϕ', λ', k^N and values V^h, V, V^1, V^k, V^e that solve the firm and household problems and clear markets as described below:

1. V, V^1, V^k and V^e solve Equations (3) - (8), and ℓ, k', b', o are the associated policy functions for incumbent firms
2. V^h solves Equation (1) with C, N, ϕ, and λ being the household's associated policy functions
3. The distribution of firms over $k \times b \times \varepsilon$ is given by μ and evolves according to $\mu' = \Gamma(\mu)$
4. The labour market clears $N = \int \ell(k, \varepsilon; S, \mu)\mu(d[k \times b \times \varepsilon)$
5. The goods market clears $C = \int \left(z\varepsilon k^\alpha \ell^\gamma + (1-\delta)k - \mathbb{I}^O[k'] - \mathbb{I}^\mathbb{N}[k^{N\prime}] \right) \mu(d[k \times b \times \varepsilon])$
6. Price q clears the bond market
7. Prices ρ_0 and ρ_1 clear the market for shares

3. Analysis of firm decision rules

Incumbent firms choose their labour demand optimally each period. This labour demand depends on the prevailing wage, their capital stock, and their output productivity as determined by aggregate and idiosyncratic productivity. As such, their optimal labour choice is $\ell = \left(\frac{w(S,\mu)}{\gamma z \varepsilon}\right)^{\frac{1}{\gamma-1}} k^{\frac{-\alpha}{\gamma-1}}$.[8]

A firm's first-best choice of capital can be obtained by substituting for ℓ in Equations (5) or (7) and maximising with respect to k'.[9] After applying the Benveniste-Scheinkman condition to replace $V(m', \varepsilon'; S', \mu')$ and using the fact that the borrowing constraint does not bind for this first-best choice of capital, the analytical expression for this is given by k^* below:

[8] The fact that this optimal expression does not contain any terms related to the earnings borrowing constraint is because firms borrow against expected *future* earnings. If, instead, firms were to borrow against their current earnings then the optimal labour choice for firms that were borrowing constrained would include an additional term as their optimum choice of labour today would directly affect how much debt they could take on.

[9] Note that this difference from Khan and Thomas (2013) is due to the fact capital in my model is completely reversible whereas their approach included irreversibility of capital. The irreversibility of capital was necessary in their approach due to their targeting of various moments related to firm investment over the business cycle. As I do not target those moments, but instead target the firm size distribution, I can follow Jo (2021) and have capital that is completely reversible (as well as base the collateral constraint on a firms future choice of capital k' rather than their current stock of capital k).

$$k'^{*}(\varepsilon,S,\mu)=\left(\frac{\frac{p(S,\mu)}{\beta}-(1-\delta_k)E_{\pi_S}p(S',\mu')}{E_{\pi_S}p(S',\mu')E_{\pi_\varepsilon}\left[\frac{-\alpha}{\gamma-1}z'\varepsilon'\left(\frac{w(S',\mu')}{\gamma z'\varepsilon'}\right)^{\frac{\gamma}{\gamma-1}}+\frac{\alpha}{\gamma-1}w(S',\mu')\left(\frac{w(S',\mu')}{\gamma z'\varepsilon'}\right)^{\frac{1}{\gamma-1}}\right]}\right)^{\frac{\gamma-1}{1-\alpha-\gamma}} \tag{9}$$

As noted in Khan and Thomas (2013), once a firm is not constrained by the borrowing constraint they will continue to choose the first-best capital. In other words, any firm that can afford to choose the first-best level of capital without violating the non-negative dividend and / or collateral constraint will choose to do so. Then, a firm's optimum debt choice in such a situation is simply to save an amount such that they will never be constrained by the borrowing constraint today or any period in the future.

As shown by D'Urso (2024), one such debt policy is for an unconstrained firm to pay zero dividends, use as much of their earnings as needed to invest, and then save any of their profits. This optimum debt choice means that an unconstrained firm would never be bound by the borrowing constraint in the future as they would have at least as much in savings as implied by the minimum-savings policy used by Khan and Thomas (2013).[10] Importantly, market clearing in the goods market means that households are indifferent between firms paying positive dividends or paying zero dividends and saving any profits. Following D'Urso (2024), this means that all firms optimally pay zero dividends.[11]

This optimum choice of debt means that both unconstrained and constrained firms pay zero dividends. In other words, all firms choose their debt and capital so that $m-k'+q(S,\mu)b'=0$ (and their choice of debt does not exceed the relevant collateral constraint). Any firm that can afford the first best choice of capital as shown by Equation (9) therefore chooses its debt according to $b'=\frac{1}{q(S,\mu)}(k'^{*}-m)$.

When a firm borrows against capital but cannot afford the first-best choice of capital without violating either the non-negative dividend constraint or the capital borrowing constraint, the firm must choose capital $k'=\frac{m}{1-q(S,\mu)\theta_k}$. For constrained firms that choose to borrow against expected future earnings, their choice of capital is determined by expression $k'=m+q(S,\mu)\theta_e E_{\pi_S}E_{\pi_\varepsilon}\max_{\ell'}[z\varepsilon k'^{\alpha}\ell'^{\gamma}-w(S,\mu')\ell']$ (via the zero dividend condition).[12] Given the choice of k' a firm's choice of debt is then $b'=\frac{1}{q(S,\mu)}(k'-m)$.

[10] The "minimum savings rule" means that once a firm starts to pay some positive dividends, it would choose its debt / savings so that it would always be able to pay positive dividends in future regardless of the aggregate state of the world or its own idiosyncratic state. As such, under such a rule, firms would choose the smallest possible level of debt that would allow them to continue paying positive dividends even in the worst case scenario.

[11] D'Urso (2024) also notes that the linearity of the firm's value function in dividends and debt means that there is still a single equilibrium even though the zero-dividend policy is only one of a number of potentially optimal savings policies.

[12] The fact that k' appears on both sides of this expression means that there is no analytical solution to this

4. Quantitative Strategy

In this Section, I set out how I calibrate the values of parameters in my model, in which one period represents one year, to match target moments. I first describe the calibration of the model's non-stochastic steady-state before setting out how the parameters governing the exogenous aggregate shocks are determined.[13]

4.1. Steady-state Calibration

I calibrate two specifications of my model in the steady-state: one that targets the firm-size distribution in 2007 (i.e. immediately before the Great Recession) and one that targets the firm-size distribution in 1980. In the 2007 calibration, there are 18 target moments and 11 (internally) calibrated parameters, so my model is overidentified. In the 1980 calibration, I keep all parameter values the same except for those parameters that govern the idiosyncratic productivity shocks; I choose these to target the firm-size distribution in 1980 (i.e. in the calibration of this second specification, there are four parameters and 10 target moments, such that its calibration is also overidentified). Two parameters are chosen outside the model: the value of β representing the representative household's rate of time preference is set to obtain a risk-free interest rate of roughly 4%, while the annual depreciation rate of physical capital is taken from Khan and Thomas (2013).

The targets and model moments for the parameters calibrated via the non-stochastic steady-state are presented in Table 1 and in Figure 2. The first column of the table contains the targeted moment and, although the parameters are all determined jointly, the second column of the table shows what parameters are most-closely associated with each target.[14] The third and fourth columns show, respectively, the source of the numerical target and the value of that target. Finally, the fifth and sixth columns show the actual moments from the specification of the model that targets the 2007 firm-size distribution (the fifth column) and the specification that targets the 1980 firm-size distribution (the sixth column).

The exponents on capital α and labour γ in the Cobb-Douglas production function are chosen to target, respectively, the aggregate capital:output ratio and labour's share of output. Household's dis-utility of working φ is set to target the proportion of time that is spent working, while the exogenous death probability π_d is set to match the average entry rate over the period 1980-2007. The relative size of entrants, and the proportion of total employment accounted for by entrants are predominantly determined by the

expression and it must be solved computationally.

[13] The specific approach and algorithms I use to solve my model are set out in Appendix B.

[14] The parameters are obtained via the Nelder-Mead optimisation algorithm using the sum of squared percentage deviations as the objective criterion.

fraction χ of aggregate physical capital with which entrants begin. The aggregate dept-to-capital ratio and the proportion of firms that borrow against earnings are determined by the value of the maximum amount firms can borrow against capital θ_k and expected future earnings θ_e.

The jointly determined parameters in my model are able to match the majority of these moments well and only slightly miss the relative size and employment contribution of entrants (both of these are relatively unimportant targets and are only included to ensure that entrants are not too large relative to incumbents).

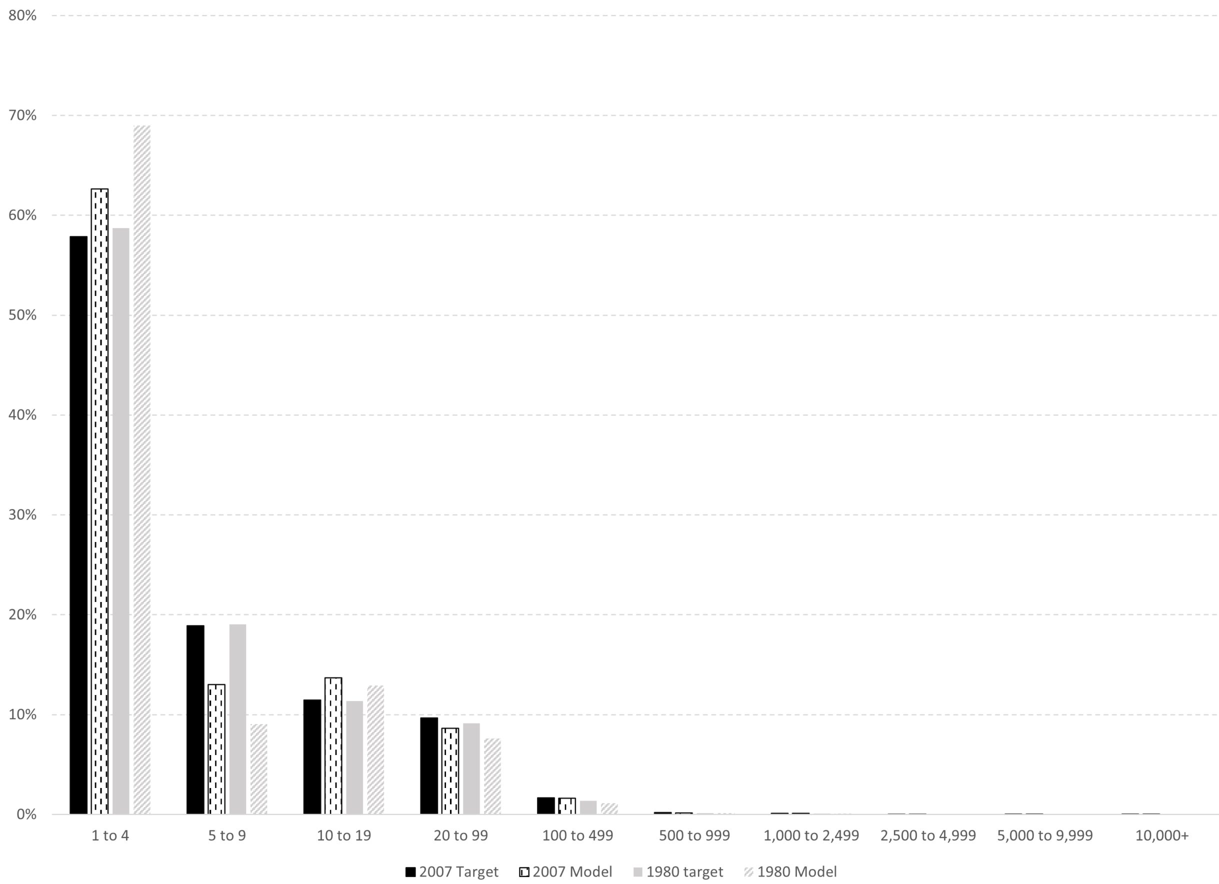
Table 1: Comparison of model specifications and targeted moments

Moment	Main parameter	Source	Data	2007 firm-size	1980 firm-size
Capital: Output ratio	Cobb-Douglas Exponent α	Literature	2.3	2.42	2.43
Labour Share	Cobb-Douglas Exponent γ	Cooley & Prescott (1995)	60%	60%	60%
% of time worked	Household disutility of working φ	Literature	33%	30%	29%
Entry rate	Exogenous death probability π_d	BDS	10.6%	10.6%	10.6%
Relative average size of entrants vs incumbents	Entrant fraction of the aggregate capital stock χ	BDS	25.6%	16.1%	16.4%
Proportion of total employment by entrants	Entrant fraction of the aggregate capital stock χ	BDS	3.0%	1.9%	1.9%
Aggregate debt-to-Capital ratio (1954-2006)	Borrowing constraints θ_k and θ_e	FRB and NIPA	0.37	0.30	0.30
Proportion of firms borrowing against earnings	Borrowing constraints θ_k and θ_e	Lian & Ma (2021)	80%	93.5%	93.6%

Also jointly determined along with the aforementioned parameters are those governing the distribution of the idiosyncratic output productivity shock. In the first instance, they are chosen to match the firm-size distribution observed just before the Great Recession; i.e. the firm-size distribution in 2007. Then, using the same values for all other parameters used to calibrate the 2007 specification, they are chosen to target the firm-size distribution seen in 1980.

As I follow Jo (2021) in using a Pareto distribution, there are four values that need to be calibrated (the upper and lower bounds of the distribution, the shape of its distribution, and the probability of a firm keeping its current productivity level). The solid bars in this graph show the proportion of firms in each employment bin in 2007 (the black bars) and 1980 (the grey bars), while the shaded bars show the proportion of firms in each bin in the calibrated steady-state of the specification targeting the 2007 distribution (the vertical dashed bar) and 1980 (the diagonally-shaded bar). In both instances, my calibration captures the Pareto distribution of firms over these employment bins relatively well, with just a few too many firms that are small relative to others.

Figure 2: Comparison of firm-size distribution targets and calibrated model results



Note: The data (the solid bars) are the BEA firm-size distributions for each of 2007 (the black bar) and 1980 (the grey bar), while the shaded bars show the results obtained in the non-stochastic steady state of the specification of the model targeting the distribution in 2007 (the vertical dashed bar) and 1980 (the diagonally shaded bar).

Table 2 contains the calibrated parameter values that are kept the same across both specifications of the model, while Table 3 shows the values of the idiosyncratic productivity shocks in each specification.[15]

To validate the model against some non-targeted moments, I examine how firm size affects whether or not firms borrow against their capital or against their earnings. In particular, I see whether or not my model can capture Lian and Ma (2021)'s finding that smaller firms tend to borrow against capital and larger firms tend to borrow against earnings. The results of this validation check are shown in Figure 3, which shows the proportion of

[15] These parameter values θ_k for capital or θ_e for expected earnings mean that only 0.004% of firms are technically insolvent (i.e. have negative cash-on-hand) in the steady-state. This means that only a very small proportion of the exiting exogenous firms cannot repay their debt when their exit, with these unpaid debts ultimately covered by the representative household.

Table 2: Parameter values common across 1980 and 2007 specifications

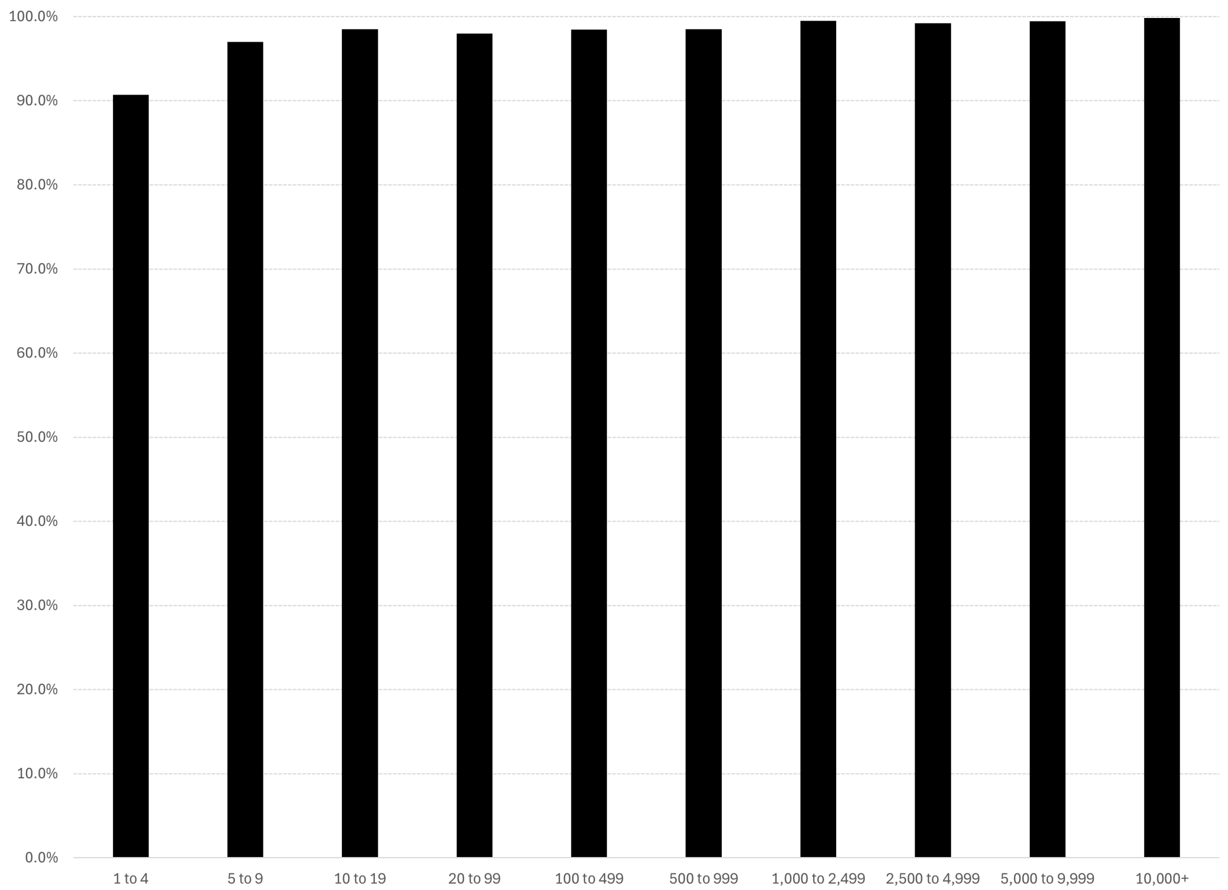
Parameter	Description	Value
α	Cobb-Douglas Exponent on capital	0.30
γ	Cobb-Douglas Exponent on labour	0.60
φ	Household disutility of working	2.38
π_d	Exogenous probability of death	0.106
χ	Entrant starting capital as % of aggregate	15.8%
θ_k	Borrowing constraint for capital as collateral	0.935
θ_e	Borrowing constraint for earnings as collateral	5.27
β	Household rate of time preference	0.96
δ	Capital depreciation rate	0.069

Table 3: Parameter values for the idiosyncratic productivity distribution

Parameter	Description	Value for 2007	Value for 1980
ε_{low}	Lowest value of ε for 2007 firm-size distribution	0.721	0.722
ε_{high}	Highest value of ε for 2007 firm-size distribution	2.33	2.37
Ψ	Pareto shape of ε for 2007 firm-size distribution	6.55	6.75
ξ	Probability of keeping current ε for 2007 firm-size distribution	0.70	0.70

firms in the model within each corresponding employment size bin that borrow against their earnings (with the remaining proportion of firms borrowing against their capital stock). Although more than 90% of firms at each employment size in the model borrow against their earnings, the proportion of firms that do so is smaller for smaller firms. For example, only 91% of firms with 1-4 employees borrow against earnings whereas almost 100% of firms whereas almost 100% of firms in the largest employment size bin borrow against their earnings. As such, my model is able to capture the pattern in the data observed by Lian and Ma (2021).

Figure 3: Proportion of firms in each employment size bin that borrow against their earnings



Note: The graph shows the proportion of firms in each size bin that borrow against their earnings. The remaining portion of firms adding up to 100% borrow against their capital stock.

4.2. *Calibration of aggregate shocks*

I assume that the aggregate productivity term z follows a mean zero AR(1) process such that $\log z' = \rho_z \log z + \eta_z'$ where η_z' is normally distributed with a mean of zero and variance of $\sigma^2_{\eta_z'}$. I estimate the persistence ρ_z and standard deviation of the shock σ_z parameters from the Solow Residual measured using Hodrick-Prescott filtered data on real annual US GDP and private capital along with total civilian employment hours created by Cociuba, Prescott and Ueberfeldt (2018) over the period 1954-2007. This process obtains a persistence $\rho_z = 0.744$ and standard deviation $\sigma_z = 0.0128$. I discretise the process across three gridpoints using the Rouwenhorst algorithm to obtain its values and transition probabilities.

I take the transition probabilities of the borrowing constraint parameter (θ_k and θ_e) from Khan and Thomas (2013). Specifically, I assume that each borrowing constraint parameter can take only two values (with its calibrated steady-state value representing "normal times" and a lower, calibrated value, representing a "financial crisis") and follows a Markov chain process represented by the matrix:

$$\Pi_{\theta_j} = \begin{bmatrix} p_0 & 1 - p_0 \\ 1 - p_l & p_l \end{bmatrix}$$

where $j = k, e$ with $p_0 = 0.9765$ and $1 - p_l = 0.3125$ set so as to achieve an average financial crisis duration of roughly 3 years for each type of collateral and ensure that the economy spends roughly 7% of the time with each collateral's borrowing constraint at its lower value.

Therefore, the transition probabilities for each credit shock θ_k and θ_e are the same. I choose the lower value of each borrowing constraint so that aggregate debt falls by roughly 26% when the economy calibrated to the 2007 firm-size distribution is subject to a financial crisis.[16] Following Ingholt (2022) and Kariya (2022)'s separate findings that the capital and earnings constraints were tight at different times during the Great Recession, I model a financial crisis as three years of the capital constraint at its lower value followed by three years of the earnings constraint at its lower value (in each case, aggregate TFP is at its median value).[17] I assume that both collateral constraints tighten by the same percentage amount during the Great Recession such that I have one target (the fall in debt) and one parameter (the percentage drop in each collateral constraint). This results in the borrowing constraints being 87.5% of their steady-state levels (i.e. $\theta_{k,low} = 0.82$ and $\theta_{e,low} = 4.61$). These same low values of the borrowing constraints are also applied to the economy that is calibrated to the 1980 firm-size distribution.

These calibrations of the different model specifications produce the business cycle moments in Table 4 for the specification calibrated to the 2007 firm-size distribution and Table 5 for the specification calibrated to the 1980 firm-size distribution. In each table, the first row shows the mean of each series, while the second shows their coefficient of variation. The third row shows the volatility of each series relative to that of output. Consistent with Drechsel (2023)'s finding that firms do not tend to change the type of collateral they use, the proportion of firms that use earnings as collateral is relatively persistent over the business cycle (its correlation with GDP is relatively low).

Note, however, although most series' relative volatilities are similar to what one would expect (e.g. consumption is less volatile than output, investment has a larger volatility relative to its own mean than do output and capital), aggregate capital K is more volatile than output in both specifications. This is predominantly due to the capital stock with which new entrants start the period varying as the aggregate capital stock itself varies. This means that fluctuations in the aggregate capital stock caused by changes in out-

[16] The 26% target for the fall in aggregate debt is obtained from Khan and Thomas (2013).

[17] Only one of the collateral constraint parameters is at its lowest value at any one time when modelling the Great Recession in Section 5, but during the simulation to obtain business cycle moments, both collateral constraint parameters can be at their lowest value at the same time.

Table 4: Simulated business cycle moments for 2007 firm-size

	Y	C	K	I	L	% firms using earnings
Mean	0.46	0.39	1.11	0.08	0.30	0.93
Coefficient of Variation	3.6%	2.4%	3.5%	13.8%	2.2%	1.1%
$\frac{\sigma_x}{\sigma_Y}$	(0.017)	0.56	2.33	0.63	0.39	0.61
Correlation with GDP	1	0.82	0.59	0.86	0.76	0.36
Auto-correlation	0.77	0.94	0.96	0.62	0.61	0.69

Note: These moments are obtained from a simulation of 35,000 periods of the specification with parameter values that target the 2007 firm-size distribution, dropping the first 500 periods of that simulation. These moments have not been filtered or smoothed. The first two rows show, respectively, the mean and coefficient of variation of each series. The third row shows the standard deviation of each series relative to that of output, except for the first column, which shows the actual standard deviation of output. The fourth row shows each series' contemporaneous correlation with output, while the last row contains each series 1-period auto-correlation.

Table 5: Simulated business cycle moments for 1980 firm-size

	Y	C	K	I	L	% firms using earnings
Mean	0.45	0.38	1.08	0.07	0.30	0.93
Coefficient of Variation	3.5%	2.4%	3.4%	13.2%	2.0%	1.1%
$\frac{\sigma_x}{\sigma_Y}$	(0.016)	0.56	2.30	0.61	0.38	0.64
Correlation with GDP	1.00	0.83	0.59	0.86	0.76	0.31
Auto-correlation	0.76	0.95	0.96	0.59	0.57	0.70

Note: These moments are obtained from a simulation of 35,000 periods of the specification with parameter values that target the 1980 firm-size distribution, dropping the first 500 periods of that simulation. These moments have not been filtered or smoothed. The first two rows show, respectively, the mean and coefficient of variation of each series. The third row shows the standard deviation of each series relative to that of output, except for the first column, which shows the actual standard deviation of output. The fourth row shows each series' contemporaneous correlation with output, while the last row contains each series 1-period auto-correlation.

put productivity or the borrowing constraint are exacerbated by this variation in new entrants' initial capital stock. All series also have the expected sign of their correlation with output and their own-auto-correlation, as shown in the fourth and fifth rows of each table.

Each of output, consumption and capital has a lower average value in the 1980 firm-size calibration compared to the 2007 calibration. Each is also (slightly) less volatile under the 1980 calibration compared to the 2007 specification. This latter aspect suggests that aggregate shocks, both financial and TFP, might have slightly smaller effects in terms of affecting aggregate variables under the 1980 firm-size distribution compared to the 2007 one. In other words, the presence of more small firms relative to larger ones could dampen the effect of aggregate shocks on the economy. This is examined in more detail in the following Section regarding the economy's aggregate response to a tightening of the collateral constraints (i.e. to a financial shock).

5. Response to Financial Shocks

In this Section I set out the model economy's recovery from a financial crisis and compare these results between the two firm-size distributions. Specifically, I examine the recovery caused by a three-period drop in the capital collateral constraint θ_k followed by a three-period drop in the earnings collateral constraint θ_e (this reflects results obtained by Ingholt (2022) and Kariya (2022) that the capital and earnings borrowing constraints tightened at different times during the 2008 Financial Crisis).[18] [19]

I find that the economy's response is slightly more muted when the initial firm size-distribution matches that observed in 1980 compared to that observed in 2007. In particular, the troughs of aggregate output, capital stock, consumption, and debt are not as deep in the 1980 calibration compared to the 2007 specification: these troughs are roughly 0.4% points higher in the former calibration. This is due to the response of smaller firms when the collateral constraints tighten: small firms actually increase their capital stock and output (relative to steady-state) in response to the tighter collateral constraints as a way of ameliorating the effect of those tighter constraints on their growth. As such, the 1980 firm-size distribution (with its higher proportion of small firms and lower proportion of large firms) has a less negative aggregate response to the financial crisis caused by the tightening of the collateral constraints.

I first present the aggregate response comparing between each specification of the firm-size distribution before examining the responses across different elements of the distribution. Although I focus on the response to financial shocks here, each economy's response to a shock to aggregate TFP is contained in Appendix A.

5.1. Aggregate response

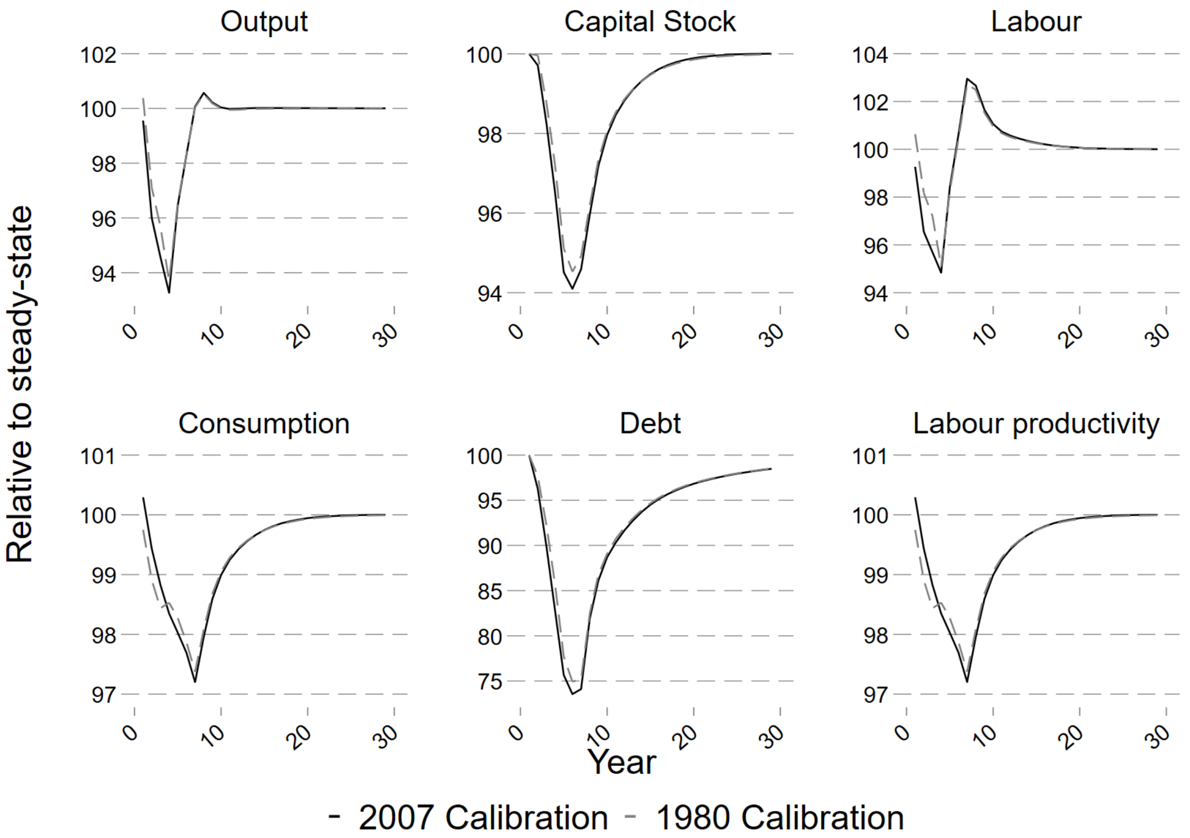
Figure 4 shows the response of aggregate variables to a financial crisis created by the aforementioned description of tightening collateral constraints. In each panel of this figure, the solid black line shows the path of the variable in the economy calibrated to the 2007 firm-size distribution, while the dashed grey line shows the response for the economy that has been calibrated to the 1980 firm-size distribution.

In each case, the financial crisis causes large drops in each of output, consumption, the capital stock and debt. Labour and labour productivity also fall during the financial crisis.

[18] For this exercise, the economy is allowed to run for 75 periods at the median aggregate productivity level and the steady-state values of the collateral multiples before the shock is applied.

[19] Note that the amount by which the collateral constraints tighten is the same for the 2007 firm-size distribution specification as it is for the 1980 firm-size specification; this is because the objective of the exercise is to compare how the change in the firm-size distribution affects the response to the financial crisis such that it would not be appropriate to re-calibrate the low values of the borrowing constraint parameters to target a 26% fall in debt under that specification.

Figure 4: Recovery of aggregate variables from a financial crisis under different firm-size distributions



Note: The graphs show the simulated time path of aggregate variables in the model economy calibrated to different firm-size distribution under the effect of tightening of the collateral constraints. Each economy is started at the steady-state distribution and runs for 75 periods with exogenous aggregate productivity at the median value of 1 and the collateral constraints at their steady-state level. Subsequently, the capital collateral constraint is at its low level for three periods before returning to its steady-state level. In the following three periods, the earnings collateral constraint is at its low level before returning to its steady-state level. The response of the economy calibrated to the 2007 firm-size distribution is shown by the solid black line, while the response for the 1980 calibration is shown by the dashed grey line.

Moreover, the timing of the trough in each variable is the same in both firm-size calibrations as each variable reaches their lowest point at the same time across both calibrations. However, the initial impact of the crisis differs across the two specifications. In particular, output is actually higher that its steady-state level in the initial period of the crisis in the 1980 firm-size calibration whereas it is below its steady-state level under the 2007 calibration. This is driven by labour being higher under the 1980 specification, such that more output can be produced with the steady-state level of capital stock. Furthermore, one period into the crisis, the capital stock has fallen under the 2007 specification but is almost unchanged under the 1980 specification. This is driven by the 1980 specification having a larger initial drop in consumption than the the 2007 specification.

The capital stock for the 1980 specification remains higher than in the 2007 specification for each year during the financial crisis and recovery from it, with its trough being 0.5% points higher in the 1980 specification. This means that output in the 1980 specification remains at least as high as it is in the 2007 calibration, with its trough being about 0.4% points higher under the 1980 calibration compared to the 2007 calibration (with a similar story for aggregate debt and labour). Hence, even though consumption in the 1980 calibration is below consumption in its 2007 counterpart for the first few periods of the crisis, that changes after four periods: from period four on, consumption in the 1980 version is always higher (relative to their respective steady-states) compared to the 2007 calibration. This is also true for the path of labour productivity.

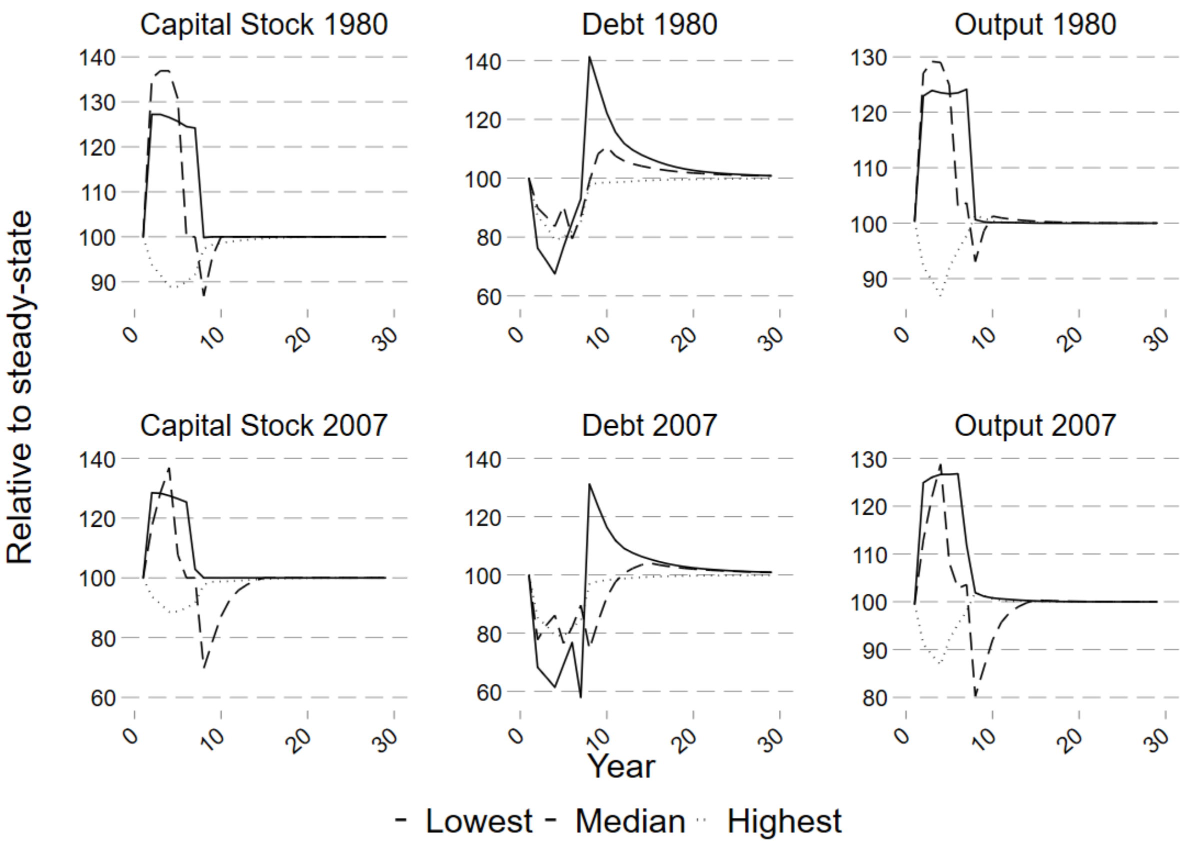
This means that the economy's response to a financial crisis caused by tightening of the collateral constraints is slightly muted when the firm-size distribution is as it was in 1980 compared to the distribution that prevailed in 2007. In other words, the presence of relatively more small firms and relatively fewer large firms seems to ameliorate the recession that results from a financial crisis.

5.2. *Response over the firm-size distribution*

The reason for the moderated response in the 1980 firm-size distribution calibration can be discerned from Figure 5 below. Specifically, this figure shows the paths of the capital stock, debt, and output for each of the bottom, median, and top quintile of firms as per their stock of capital. The solid lines show the paths of each variable for the firms in the lowest quintile of capital stock; the dashed lines show this for firms in the median quintile of capital stock; and the dotted lines do likewise for firms in the top quintile of the capital stock. The top row of panels shows the paths for firms under the 1980 calibration while the bottom row of panels contain the paths for firms in the 1980 specification.

Over the path of the financial crisis, small and medium-sized firms actually increase their capital stock and their output, whereas large firms reduce those two variables. This is because small and medium-sized firms are still affected by the collateral constraint, such that a tightening of the collateral constraint substantially reduces how much they can borrow and how quickly they can accumulate capital. As such, these firms increase their capital and output to try to ameliorate the effect of the tighter collateral constraints on their growth paths, such that the debt held by these firms does not reduce by as much as it would have done otherwise. This reinforces Luttmer (2011)'s finding that small firms tend to grow more quickly than larger firms as these smaller firms grow even after a financial shock, whereas larger firms contract after such a shock. My results are also consistent with Fort et al. (2013)'s result that smaller firms respond more to shocks than do larger firms (note that in the results above the peak absolute magnitude of the change in the lowest quintile firms' capital stock is roughly 30% whereas it is only 10% for firms

Figure 5: Recovery of aggregate variables from a financial crisis under different firm-size distributions, split by quintile of capital



Note: The graphs show the simulated time path of aggregate variables in the model economy calibrated to different firm-size distribution under the effect of tightening of the collateral constraints. Each economy is started at the steady-state distribution and runs for 75 periods with exogenous aggregate productivity at the median value of 1 and the collateral constraints at their steady-state level. Subsequently, the capital collateral constraint is at its low level for three periods before returning to its steady-state level. In the following three periods, the earnings collateral constraint is at its low level before returning to its steady-state level. The response of the economy calibrated to the 2007 firm-size distribution is shown in the bottom row of panels, while the response for the 1980 calibration is shown by the top row of panels. The solid lines show the response for the lowest capital quintile in each calibration; the dashed lines the median quintile; and the dotted lines the highest quintile.

in the highest quintile).[20] [21]

This means that the 1980 firm-size distribution, with a higher proportion of small firms

[20] It is important to note that Fort et al. (2013)'s empirical approach allows them to distinguish between a firm's age and a firm's size, such that they can disentangle different effects of firm age and firm size in their results. This helps to drive their finding that small firms are more sensitive to the business cycle than are larger firms because they include old and middle-aged firms in their group of small firms. In my model, however, all small firms are also young (I do not track a firm's age as part of a firm's state) such that there are no old small firms. This means that Fort et al. (2013)'s results are not directly comparable to the results I present here even though my results are consistent with theirs in terms of the magnitude of responses across firm size.

[21] My results are not directly comparable with the empirical findings of Crouzet and Mehrotra (2020) for two reasons: first, they split firms by size by defining large firms as just those in the top 1% of assets and small firms as those in the bottom 99% of assets, whereas I use a more graduated size distinction with small firms being only those in the bottom 20% (i.e. some of the large firms in my model would be classified as small firms in their study); and second, they examine the response of firms to all shocks to the US economy since 1977, whereas I focus solely on the response to the financial shock that resulted in the Great Recession.

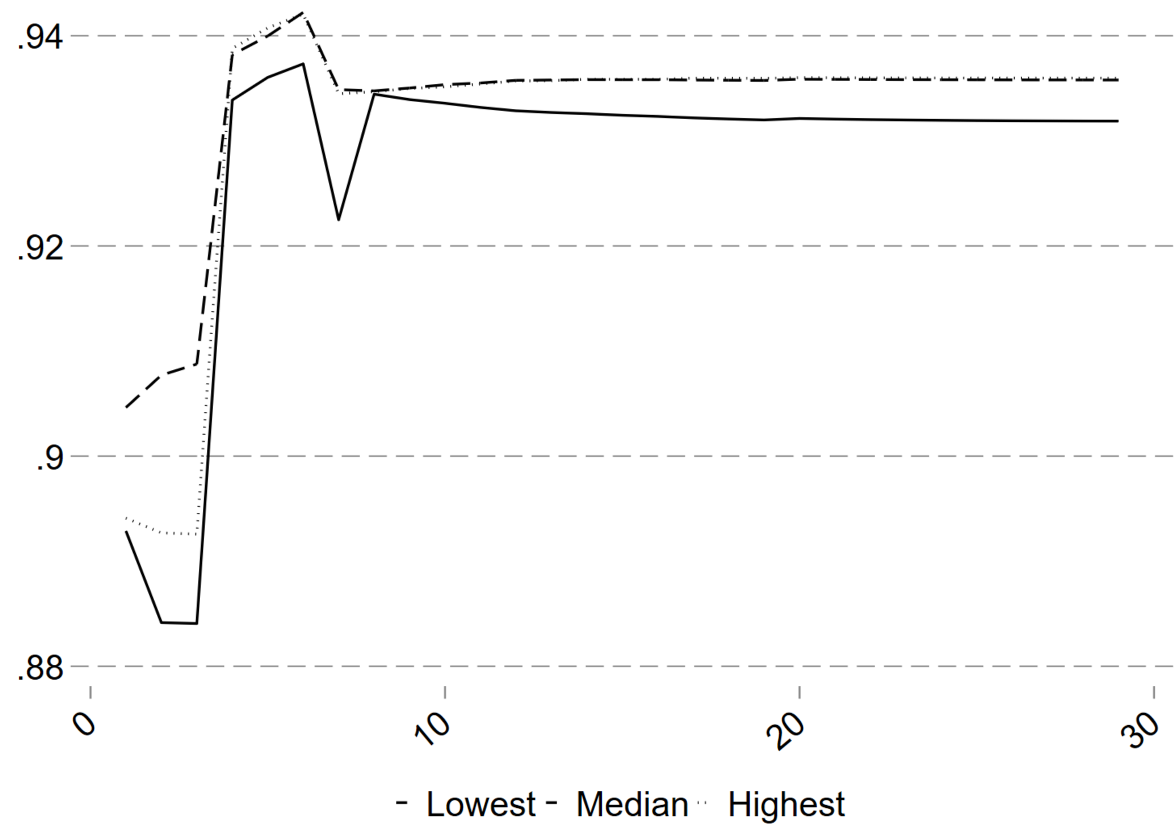
and a lower proportion of larger firms, does not experience as large a drop in their capital stock or output as observed when the same shocks are applied to the 2007 calibration. However, as larger firms account for the majority of the aggregate capital stock, output, and debt, the overall path of the economy's response remains driven by the response of these larger firms. Hence, the aggregate response of these variables is a large drop in output and the capital stock that is just ameliorated by the presence of more small firms and fewer large firms in the 1980 firm-size distribution compared to the 2007 distribution.

Focusing on just the effect of the financial crisis on the 2007 firm-size distribution 9the results for the 1980 distribution are very similar), Figure 6 shows how the proportion of firms that borrow against earnings changes during the simulated financial crisis and recovery from it, split by firm size. As before, the solid line shows the paths of each variable for the firms in the lowest quintile of capital stock; the dashed line shows this for firms in the median quintile of capital stock; and the dotted line does likewise for firms in the top quintile of the capital stock. It is clear that each size of firm follows a very similar pattern: in the first three periods (when the capital borrowing constraint tightens) firms actually switch away from borrowing against earnings towards borrowing against capital. Then, from periods four to six (when the capital borrowing constraint returns to normal and the earnings borrowing constraint tightens), firms then choose to switch back to borrowing against earnings slightly more than in the steady-state. Once the collateral constraint shocks have both returned to normal (from period seven on), the proportion of firms borrowing against earnings quickly returns to its steady-state level.

The fact that firms switch towards borrowing against capital when the capital constraint tightens might seem counterintuitive, but can be explained by the following factors. First, when the capital constraint tightens at the start of the crisis, small and medium firms increase their capital stock and therefore can borrow (relatively) more against capital and grow more quickly than if they borrowed against earnings (due to the presence of diminishing marginal returns to capital). Hence, some small and medium firms switch to borrowing against capital when the capital constraint is tightened.

Second, affecting all firms, and the main driving force as to why some large firms also switch towards borrowing against capital is the presence of the non-negative dividend constraint. This constraint means that firms must always ensure that they can repay their debt next period. If firms borrow against their choice of capital, then they know for certain that they can pay back their debt next period as their capital level is guaranteed (and their debt is less than the undepreciated level of capital, by definition). However, if firms borrow against their expected earnings, then there is a chance that those earnings next period might be below their expected level such that firms do not have sufficient funds to repay their debts. As such, the increased amount of uncertainty as to a firm's expectations of their earnings when the capital constraint tightens means that some firms

Figure 6: Proportion of firms that borrow against earnings during the recovery from a financial crisis, split by quintile of capital



Note: The graph shows the simulated time path the proportion of firms that borrow against earnings in the model economy calibrated to the 2007 firm-size distribution under the effect of tightening of the collateral constraints. Each economy is started at the steady-state distribution and runs for 75 periods with exogenous aggregate productivity at the median value of 1 and the collateral constraints at their steady-state level. Subsequently, the capital collateral constraint is at its low level for three periods before returning to its steady-state level. In the following three periods, the earnings collateral constraint is at its low level before returning to its steady-state level. The solid lines show the response for the lowest capital quintile in each calibration; the dashed lines the median quintile; and the dotted lines the highest quintile.

switch to borrowing against the more certain form of collateral (capital). In other words, firms are responding in a precautionary way when the capital constraint tightens.

Conversely, when the capital constraint returns to normal and the earnings constraint tightens, we see the opposite effect: slightly more firms borrow against earnings relative to the steady-state. Again, this is driven by firms acting precautionarily so as to ensure that they always have sufficient funds such that they do not violate the non-negative dividend constraint. As the proportion of firms borrowing against earnings when the earnings constraint is tightened is relatively similar to the steady-state proportion of firms that borrow against earnings, this proportion then rapidly returns to the steady-state level when both borrowing constraints are returned to their steady-state value.

6. Contribution of each collateral constraint

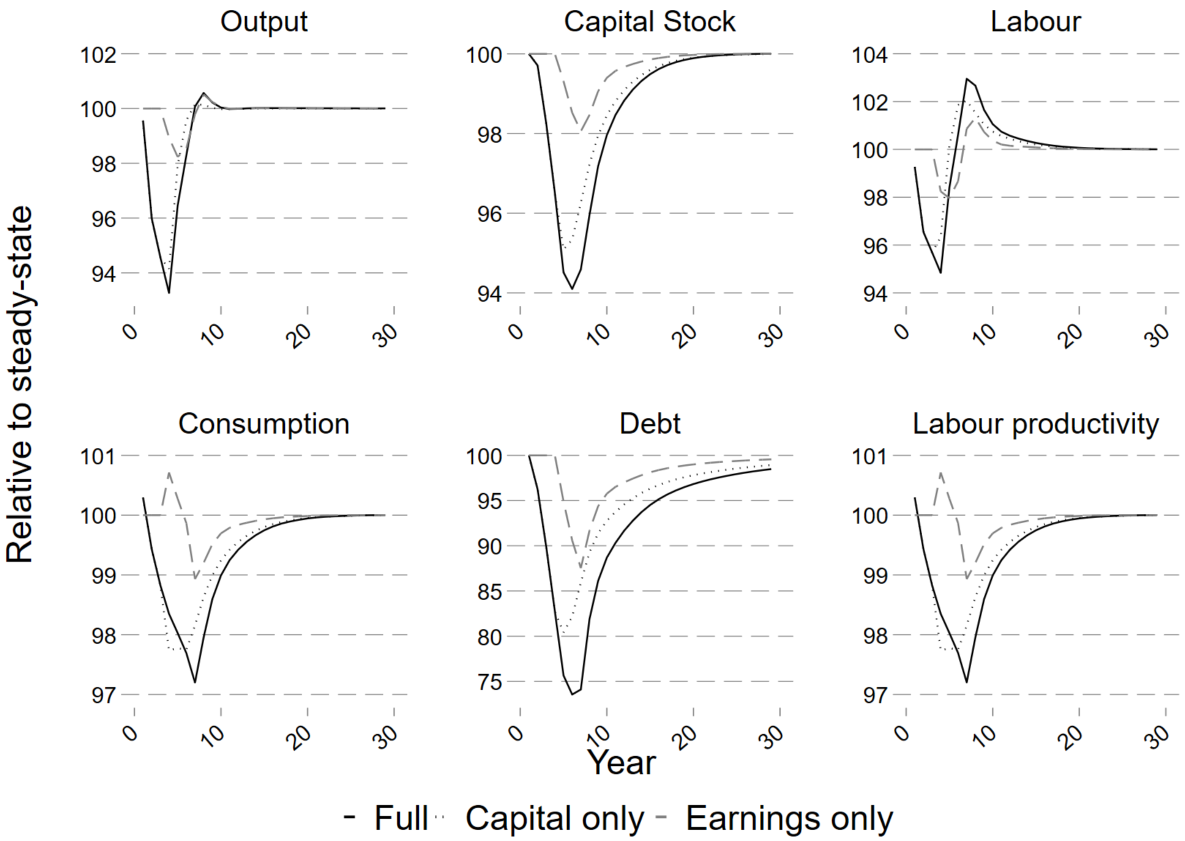
I am also able to examine the contribution of each collateral constraint (capital and earnings) to the overall path of the economy when both collateral constraints are tightened, using just the 2007 firm-size distribution calibration. In particular, I compare between three simulated recoveries as shown in Figure 7 : 1) as in the previous Section, a three-period drop in the capital collateral constraint θ_k followed by a three-period drop in the earnings collateral constraint θ_e (the solid line); 2) a three period drop in just the capital collateral constraint while the earnings constraint is kept at its steady-state level throughout (the dotted line) ; and 3) three periods of both collateral constraints at their steady-state level followed by a three-period drop in the earnings collateral constraint (the dashed line). In this way the timing of each shock hitting is exactly the same across the three simulated recoveries.

As one would expect given the timing of the shocks, the time paths for the full shocks and the capital only shock overlap completely for the first three periods. Once the earnings shock also hits in period four, its impact can be discerned by comparing the solid black line (the scenario in which both collateral constraints are tightened consecutively) to the dotted line (in which just the capital collateral constraint is tightened). For example, the additional impact of the earnings shock on output is relatively small as it reduces output by only about one percentage point compared to the reduction caused by the capital collateral constraint tightening.

Indeed, the earnings shock is applied just by itself, output falls by about two percent relative to steady-state, suggesting that the combination of the two shocks ameliorates each other when it comes to output. This is also true for some part of the path of consumption, with consumption actually being higher when the earnings collateral shock hits in period four compared to just the application of the capital collateral shock alone. However, the earnings collateral shock then exacerbates the drop in consumption after a few more periods (with labour productivity following a similar path).

The combination of both shocks also exacerbates the drop in the aggregate stock of capital and debt, albeit with the tightening of the capital collateral constraint contributing more to the ultimate trough of both variables than does the tightening of the earnings collateral constraint. This suggests that even though substantially fewer firms borrow against capital compared to earnings, the tightening of the capital collateral constraint has a disproportionately large effect on those firms, such that the path of the economy's recovery after a financial crisis caused by a tightening of both collateral constraints is predominantly driven by the effect of the initial tightening of the capital collateral constraint. Moreover, the subsequent tightening of the earnings collateral constraint does not substantially lengthen the time taken for the economy to recover to its pre-crisis level.

Figure 7: Comparison of effect of the different collateral shocks



Note: The graphs show the simulated time path of aggregate variables in the model economy calibrated to the 2007 firm-size distribution under the effect of different timings of the collateral constraints tightening. Each economy is started at the steady-state distribution and runs for 75 periods with exogenous aggregate productivity at the median value of 1. The solid black line shows the recovery of the economy when subjected to both shocks (a three-period drop in the capital collateral constraint and then a three-period drop in the earnings collateral constraint); the black dotted line shows the economy's recovery after it is subjected to just a three-period drop in the capital collateral constraint and the earnings collateral constraint is kept at its steady-state value throughout; and the grey dashed line shows the recovery when the economy has three more periods at the steady-state before the earnings collateral constraint is tightened for three periods and the capital collateral constraint is kept at its steady-state level.

Indeed, this latter aspect of a more limited effect of the tightening the earnings borrowing constraint is consistent with how the proportion of firms borrowing against earnings changes over the crisis when both borrowing constraints are tightened consecutively. In particular, and as discussed in Section 5.2, the proportion of firms that borrow against earnings is relatively similar to the steady-state level of that proportion when the earnings borrowing constraint is tightened, whereas when the capital borrowing constraint is tightened there is a (relatively) large change in this proportion..

As the troughs of both output and consumption when both shocks hits closely matches those seen for each variable in the recovery from the 2007 / 2008 Financial Crisis, and it is the tightening of the capital collateral constraint that accounts for most of that trough in the model, it suggests that the capital collateral constraint is able to capture a lot of the US economy's recovery from 2008 on. Hence, further to Shah (2024), it appears that even though tightening the earnings collateral constraint leads to a different path of the recovery compared to tightening the capital collateral constraint, a capital collateral constraint is able to capture a substantial portion of the US economy's actual recovery path despite the majority of firms borrowing against their earnings rather than against their capital.

7. Conclusion

I have shown that the economy's response to a financial crisis depends on the distribution of firms over their size. Using a discrete time dynamic stochastic general equilibrium model with heterogeneous firms that hold physical capital and debt subject to a borrowing constraint and can choose whether they borrow against their future expected earnings or against their capital, I calibrate the non-stochastic steady-state to match the firm-size distribution observed in either 1980 or 2007. The distribution of firms in 1980 has relatively more small firms and relatively fewer large firms compared to 2007.

For each calibration of the model, I simulate the response to a financial crisis that takes the form of a three-period tightening of the capital collateral constraint followed by a three-period tightening of the earnings collateral constraint, reflecting the literature regarding the timing of different collateral constraints binding. I find that although the overall path of aggregates such as output and consumption are similar regardless of the firm-size distribution, the 1980 calibration has a less deep trough in each of these variables. This is because smaller firms actually increase their capital stock and output when the collateral constraints are tighter in an attempt to ameliorate the effect of those tighter constraints on the amounts they can borrow (and, hence, how quickly they can grow). As such, a distribution with a higher proportion of smaller firms experiences has a more muted response to a negative shock to collateral constraints. However, as large firms still account for the majority of output and capital stock, and these variables fall for these firms, the economy's overall path is one of a large drop in aggregates.

This result suggests that if a policymaker is worried about the possibility of a financial crisis happening and wishes to reduce its potential effects, the policymaker might want to focus on encouraging more small firms to continue producing while at the same time potentially discouraging large firms from forming (or from being too prevalent in the economy).

Moreover, I find that the majority of the aggregate economy's response to the sequential tightening of collateral constraints comes from its response to the capital collateral constraint tightening, despite the fact that the majority of firms borrow against (expected future) earnings rather than against capital. This implies that focusing solely on capital collateral constraints could be a reasonable approach to modelling the impact of a financial crisis and capturing the main aspects of the 2007 / 2008 Great Recession despite the observation of firms also borrowing against earnings.

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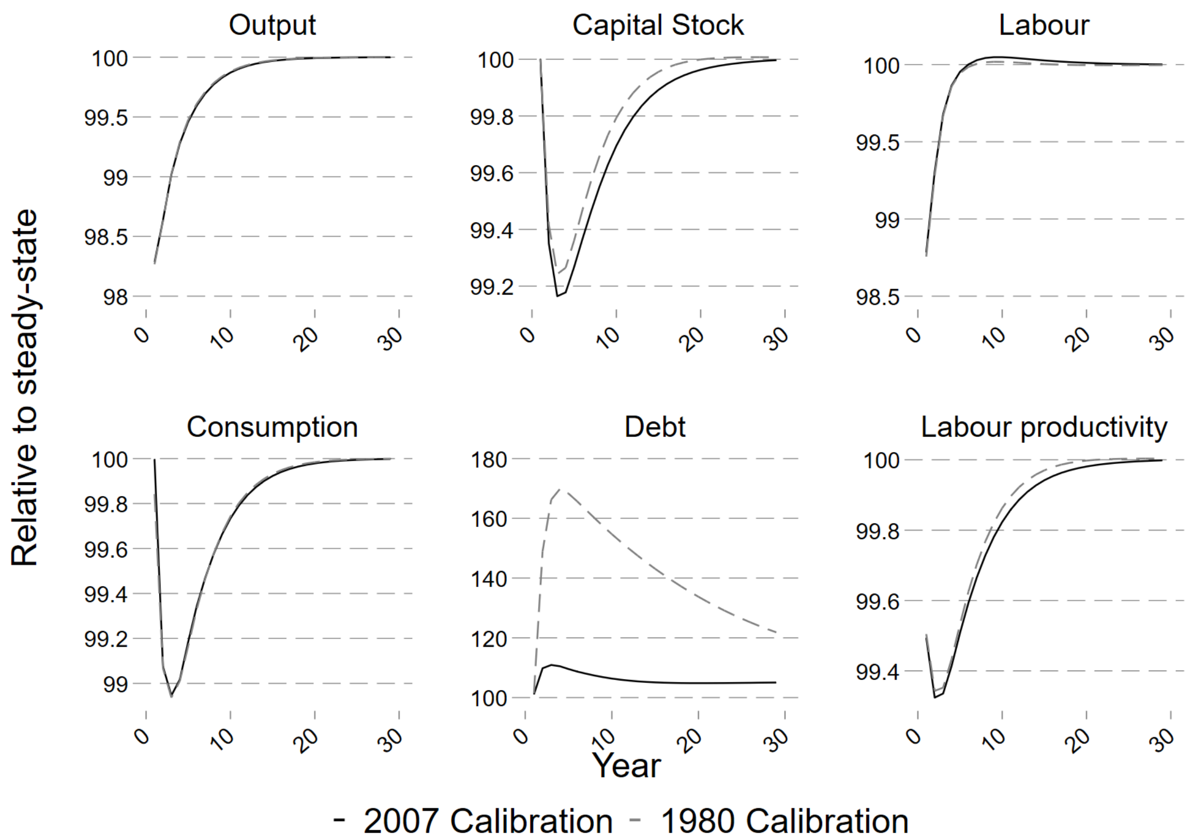
Appendix A. Aggregate Response to a TFP Shock

In this Appendix Section I set out the response of each calibrated specification of the model to a 1% drop in aggregate TFP. Specifically, the economy is allowed to run for 75 periods at the median aggregate productivity level and the steady-state value of the collateral multiples before aggregate TFP drops by 1%. Subsequently, aggregate TFP gradually returns to its steady-state level as determined by its persistence ρ_z. In each panel of Figure A.8, the solid black line shows the path of the variable in the economy calibrated to the 2007 firm-size distribution, while the dashed grey line shows the response for the economy that has been calibrated to the 1980 firm-size distribution.

Figure A.8: Recovery of aggregate variables from an aggregate TFP shock under different firm-size distributions



Note: The graphs show the simulated time path of aggregate variables in the model economy calibrated to different firm-size distribution under the effect of tightening of the collateral constraints. Each economy is started at the steady-state distribution and runs for 75 periods with exogenous aggregate productivity at the median value of 1 and the collateral constraints at their steady-state level. Subsequently, Aggregate TFP drops by 1% before gradually returning to its steady-state level of 1. The response of the economy calibrated to the 2007 firm-size distribution is shown by the solid black line, while the response for the 1980 calibration is shown by the dashed grey line.

The path of most variables is similar across the two different calibrations. In particular, output, labour, and consumption are almost identical across the 2007 and 1980 calibrations. The largest difference between the two calibrations is in terms of debt: debt reaches a peak roughly 10% above the steady-state level in the 2007 calibration whereas that peak is about 70% above the steady-state level in the 1980 specification. This large increase in aggregate debt means that the capital stock does not fall as much in the 1980 calibration compared to the 2007 specification, which also leads to labour productivity in the 1980 specification being above that of the 2007 calibration.

Overall, however, the negligible differences in the paths of output, labour, and consumption between the two calibrations suggests that differences in the firm-size distribution do not make a substantial difference to most aspects of an economy's response to an aggregate TFP shock.

Appendix B. Solving the model

In this Appendix Section I set out the methods I use to solve for the non-stochastic steady-state and the models with aggregate uncertainty.

Appendix B.1. Non-stochastic steady-state solution algorithm

In order to solve the model I discretise the output productivity shock ε using the Rouwenhorst algorithm with 13 gridpoints. I guess an aggregate level of physical capital $\bar{K}$ (which is necessary to determine the size at which new entrants start) and a consumption C, the latter implying a wage w through the household's labour-leisure condition. I then use the implied decision rules for k', o, and b' described in the main text, as well as the distribution of new entrants alongside exogenous exits, to find the stationary distribution, which is then used to obtain values for implied $\bar{K}$ and C. The initial guesses of these variables are then updating using Broyden's algorithm.

Appendix B.2. Solution algorithm for aggregate uncertainty

The presence of aggregate uncertainty in the form of varying aggregate productivity z (which can take three values) and θ_k or θ_e (each of which can take two values) means that I use the Krusell-Smith solution method to solve my model, adapted for heterogeneous firms as per Khan and Thomas (2013). This means that I approximate the distribution of firms μ using the summary statistic of the aggregate level of capital $\bar{K}$ and use that in the forecasting rules that take the form

$$\log x = \beta_{0,i} + \beta_{1,i} \log \bar{K}_t + \beta_{2,i}\xi_{e,t} + \beta_{3,i}\xi_{k,t} + v_i$$

where x is the variable to be forecast, β_j are coefficients to be estimated, $\xi_{e,t}$ and $\xi_{k,t}$ are dummy variables taking the value 1 if there was a low value of, respectively, θ_e or θ_k one period ago, v is the error term, and i represents the level of aggregate productivity and the borrowing constraints in the current period. Each forecast rule is obtained via regressions using the final 34,500 observations from a simulation of 35,000 periods (i.e. the first 500 periods of the simulation are dropped when estimating the forecast rules).

I require forecasting rules for two variables: the level of aggregate capital next period $\bar{K}'$ (to determine the expected level of the wage next period and, therefore, firms' optimum choices); and the household's marginal utility this period p (to determine the marginal utility value of firm profits today, as firms are owned by the household). The estimated coefficients, and some regression statistics for each of these forecast rules for the specification of the model with the capital borrowing constraint are shown in Tables B.6 and B.7. The first three columns indicate to which aggregate state in the current period the forecast rule applies. Columns 4 and 5 contain the estimated coefficient values, while column six shows the standard error for the regression. The final column presents the R^2 of the regressions and shows that it is above or close to 0.99 for all regressions.

Table B.6: Aggregate Capital forecast rule for the 2007 firm-size specification

Aggregate productivity	Capital Borrowing Constraint	Earnings Borrowing Constraint	β_0	β_1	β_2	β_3	SE	R^2
z_1	θ_k	θ_e	0.009	0.784	-6.46E-04	-1.58E-02	1.51E-03	0.996
z_1	θ_k	$\theta_{e,low}$	0.005	0.766	-7.58E-04	-1.49E-02	1.67E-03	0.995
z_1	$\theta_{k,low}$	θ_e	0.003	0.830	-3.46E-04	-1.47E-02	3.55E-03	0.991
z_1	$\theta_{k,low}$	$\theta_{e,low}$	-0.0001	0.806	1.35E-03	-1.44E-02	3.43E-03	0.994
z_2	θ_k	θ_e	0.023	0.781	-7.67E-04	-1.61E-02	1.48E-03	0.996
z_2	θ_k	$\theta_{e,low}$	0.020	0.759	-9.62E-04	-1.52E-02	1.64E-03	0.995
z_2	$\theta_{k,low}$	θ_e	0.015	0.833	-3.85E-03	-1.49E-02	7.58E-03	0.960
z_2	$\theta_{k,low}$	$\theta_{e,low}$	0.014	0.798	1.07E-03	-1.47E-02	3.87E-03	0.982
z_3	θ_k	θ_e	0.037	0.781	-5.33E-04	-1.61E-02	1.58E-03	0.996
z_3	θ_k	$\theta_{e,low}$	0.034	0.764	-1.20E-03	-1.20E-02	3.07E-03	0.980
z_3	$\theta_{k,low}$	θ_e	0.029	0.825	1.85E-03	-1.53E-02	3.60E-03	0.990
z_3	$\theta_{k,low}$	$\theta_{e,low}$	0.028	0.801	-9.06E-04	-1.40E-02	3.38E-03	0.992

Note: SE is the standard error in each regression

Similarly, Tables B.8 and B.9 show the solved forecast rules for the specification with the expected earnings borrowing constraint. As before, the regression standard errors are very small and the R^2 is above or very close to 0.99 in all cases.

Table B.7: Marginal Utility forecast rule for the 2007 firm-size specification

Aggregate productivity	Capital Borrowing Constraint	Earnings Borrowing Constraint	β_0	β_1	β_2	β_3	SE	R^2
z_1	θ_k	θ_e	1.015	-0.487	7.47E-04	4.55E-03	3.58E-04	0.999
z_1	θ_k	$\theta_{e,low}$	1.010	-0.497	7.77E-04	4.77E-03	3.07E-04	1.000
z_1	$\theta_{k,low}$	θ_e	1.012	-0.479	5.33E-04	4.64E-03	7.37E-04	0.999
z_1	$\theta_{k,low}$	$\theta_{e,low}$	1.009	-0.491	2.14E-04	4.66E-03	5.74E-04	0.999
z_2	θ_k	θ_e	1.000	-0.482	7.70E-04	4.54E-03	3.41E-04	0.999
z_2	θ_k	$\theta_{e,low}$	0.996	-0.492	7.85E-04	4.69E-03	3.18E-04	1.000
z_2	$\theta_{k,low}$	θ_e	0.996	-0.466	2.13E-03	4.73E-03	2.80E-03	0.980
z_2	$\theta_{k,low}$	$\theta_{e,low}$	0.995	-0.485	3.06E-04	4.70E-03	6.61E-04	0.998
z_3	θ_k	θ_e	0.985	-0.476	8.11E-04	4.53E-03	4.42E-04	0.999
z_3	θ_k	$\theta_{e,low}$	0.981	-0.488	8.16E-04	3.78E-03	3.59E-04	0.999
z_3	$\theta_{k,low}$	θ_e	0.981	-0.468	1.60E-04	4.59E-03	6.99E-04	0.999
z_3	$\theta_{k,low}$	$\theta_{e,low}$	0.979	-0.480	6.48E-04	4.59E-03	5.78E-04	0.999

Note: SE is the standard error in each regression

Table B.8: Aggregate Capital forecast rule for the 1980 firm-size specification

Aggregate productivity	Capital Borrowing Constraint	Earnings Borrowing Constraint	β_0	β_1	β_2	β_3	SE	R^2
z_1	θ_k	θ_e	0.003	0.787	-6.82E-04	-1.70E-02	1.50E-03	0.996
z_1	θ_k	$\theta_{e,low}$	-0.002	0.760	-6.74E-04	-1.62E-02	1.74E-03	0.994
z_1	$\theta_{k,low}$	θ_e	0.000	0.834	-2.42E-04	-1.68E-02	3.80E-03	0.988
z_1	$\theta_{k,low}$	$\theta_{e,low}$	-0.004	0.816	1.42E-03	-1.56E-02	3.43E-03	0.992
z_2	θ_k	θ_e	0.017	0.784	-8.18E-04	-1.73E-02	1.46E-03	0.996
z_2	θ_k	$\theta_{e,low}$	0.013	0.751	-9.06E-04	-1.66E-02	1.70E-03	0.994
z_2	$\theta_{k,low}$	θ_e	0.012	0.841	-4.23E-03	-1.70E-02	7.85E-03	0.948
z_2	$\theta_{k,low}$	$\theta_{e,low}$	0.010	0.801	1.15E-03	-1.59E-02	3.75E-03	0.979
z_3	θ_k	θ_e	0.031	0.784	-5.83E-04	-1.73E-02	1.57E-03	0.995
z_3	θ_k	$\theta_{e,low}$	0.027	0.755	-1.21E-03	-1.33E-02	2.96E-03	0.979
z_3	$\theta_{k,low}$	θ_e	0.026	0.828	2.18E-03	-1.74E-02	3.69E-03	0.987
z_3	$\theta_{k,low}$	$\theta_{e,low}$	0.023	0.811	-1.14E-03	-1.52E-02	3.51E-03	0.989

Note: SE is the standard error in each regression

Table B.9: Marginal Utility forecast rule for the 1980 firm-size specification

Aggregate productivity	Capital Borrowing Constraint	Earnings Borrowing Constraint	β_0	β_1	β_2	β_3	SE	R^2
z_1	θ_k	θ_e	1.021	-0.487	9.44E-04	7.08E-03	3.51E-04	0.999
z_1	θ_k	$\theta_{e,low}$	1.016	-0.489	9.41E-04	7.30E-03	2.98E-04	1.000
z_1	$\theta_{k,low}$	θ_e	1.022	-0.478	4.92E-04	7.26E-03	7.29E-04	0.999
z_1	$\theta_{k,low}$	$\theta_{e,low}$	1.019	-0.490	7.48E-05	6.93E-03	5.79E-04	0.999
z_2	θ_k	θ_e	1.007	-0.481	9.91E-04	7.09E-03	3.28E-04	0.999
z_2	θ_k	$\theta_{e,low}$	1.001	-0.481	9.75E-04	7.24E-03	3.02E-04	1.000
z_2	$\theta_{k,low}$	θ_e	1.007	-0.460	3.07E-03	7.38E-03	2.85E-03	0.975
z_2	$\theta_{k,low}$	$\theta_{e,low}$	1.005	-0.485	1.72E-04	7.01E-03	6.56E-04	0.998
z_3	θ_k	θ_e	0.991	-0.476	1.03E-03	7.07E-03	4.22E-04	0.999
z_3	θ_k	$\theta_{e,low}$	0.986	-0.478	1.05E-03	5.76E-03	3.48E-04	0.999
z_3	$\theta_{k,low}$	θ_e	0.993	-0.466	-2.29E-04	7.27E-03	6.98E-04	0.998
z_3	$\theta_{k,low}$	$\theta_{e,low}$	0.989	-0.482	8.19E-04	6.78E-03	6.07E-04	0.999

Note: SE is the standard error in each regression